

Call for Expressions of Interest	
<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Peering inside Neural Nets – What is it thinking?
<i>Company:</i>	N/A
Contact Details	
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Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Machine learning
Project Background Information	
Neural nets often produce very good performance on the datasets that are provided for their training/testing. However, it can then be very unclear how they perform or what they do in other regions of the feature space, and generally very difficult to visualise or explain how they do what they do. In particular, it could well be that in some regions of the feature space an inspection by a human would cast doubt on the validity (similar to the hallucinations that are common with LLMs). Hence, it would be useful to have a good visualisation system for experts in some domain, to be able to inspect the general structure of classification results produced from some trained neural net, and so to also be able to view how the internals of the net build to the final answer. For example, being able to visualise how a node, or multiple nodes, respond to varying the inputs. Also, for cases where the net performs less well than hoped, it could be useful to be able to identify what is happening internally. The intention is that such a system would be fall in the area of “Explainable AI” and so be able to support cases when (due to GDPR or other reasons) an explanation is needed.	

¹ An overview of different research areas in the school of computer science can be found [here](#).

Requirements	
<i>The group must provide:</i> A software system that can take a neural net (possibly trained within the system or just loaded from outside) and provide a user level view of the structure. The system should support users to select a node, or sets of nodes, and obtaining visualisations of how those nodes respond to different inputs to the net.	
Skills	
<i>Highly desirable:</i> - Basic knowledge of the workings of neural nets - Programming skills in a suitable language - Ability to do some graphics	<i>Desirable:</i> - Interest in machine learning - Interest and experience in visualisation - Some basic mathematical skills – simple linear algebra
Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.